MA342 Project 5

Iayan Ansari, Ben Egelseer, Luke Gaston, and Jack Silkaitis

May 2026

1 Introduction

Atomic structures are the fundamental building blocks of the universe, composing all physical structures across our planet, the solar system, and the universe. Understanding these interactions can provide insight into material behavior in fields such as thermodynamics, condensed matter physics, and electrical engineering.

Our model approaches this by predicting the movement and behavior of a two-dimensional atomic structure under various properties and initial conditions using a linear regression model. With this system, we provide a computed mean of properties of various materials based on their atomic structure. The methodology of this report is derived from Chapter 12 of Holder and Eichholz [1].

2 Mathematical Background

In atomic structures, entropy, the natural tendency of atoms to move from a structured to an unorganized pattern, and energy, which describes the interaction strength between neighboring atomic spins, have a competing relationship between them that can be related mathematically. The relation used in this model is the Gibbs distribution under a discrete probability distribution p. The entropy associated with a discrete probability distribution p is:

$$-\sum_i p_i \ln(p_i)$$

with $p_i > 0$ and $\sum_i p_i = e^T p = 1$.

Using the Ising model, the energy of the system can then be defined as

$$E(\sigma) = -J \sum_{(i,j)\in\mathcal{A}} \sigma_i \sigma_j - H \sum_i \sigma_i.$$

The variable σ contains the position's spin, indicated by a 1 for positive spin and -1 for negative spin. $\mathcal{A}$ represents the collection of nearest, i.e., adjacent, neighbors. This is defined by the type of lattice being modeled such as rectangular, triangular or hexagonal. We also use a periodic boundary condition, meaning the states 'above' the top row of the two-dimensional model are the bottom row, and the states to the 'right' of the last column are those of the first column and vice versa.

Since we model a ferromagnetic material, we assume $J = k = 1$, thus setting a thermodynamic temperature scale k is the Boltzmann constant, and we assume $k = 1$. We also assume $J = \eta = 1$. To capture the assumption that our Ising model is in the absence of an external field, we assume $H = 0$. The value for J could be experimentally customized to better represent a specific material but that is outside the scope of this report. The equation for the energy of the system can then be simplified into the following:

$$E(\sigma) = -\sum_{(i,j)\in\mathcal{A}} \sigma_i \sigma_j.$$

Instead of restricting the model to tiny lattices for which all states can be determined, we use larger models and approximate desired quantities by sampling the state space. Because ferromagnetic materials seek to have the same spins in low energy states, the Gibbs probability needs to be high if the energy is low. The Gibbs distribution already does this and thus the sample average is representative of the entire lattice. This would be calculated as a summation over the sampled states, shown in the following equation.

$$\bar{E} = \frac{1}{N} \sum_{i=1}^{N} E(\sigma_i) \approx \langle E \rangle.$$

To implement this in the mathematical model, an iterative process is used across the lattice to determine if a single spin should flip or stay the same. This is done primarily by determining the energy states associated with each spin configuration. Let σ^k be the current state of the lattice and $\hat{\sigma}$ be different by a single spin. The equation to determine which state proceeds to the next iteration can then be determined by

$$P(\sigma^k, \hat{\sigma}, T) = \begin{cases} 1, & E(\hat{\sigma}) < E(\sigma^k) \\ e^{(E(\sigma^k) - E(\hat{\sigma}))/T}, & E(\hat{\sigma}) \geq E(\sigma^k). \end{cases}$$

If $P(\sigma^k, \hat{\sigma}, T) \geq y$ with y being a sample of $Y \sim U(0, 1)$, then the reversed state $\hat{\sigma}$ is accepted. During each iteration, 60% of the lattice spins are randomly selected as candidates for potential spin reversal.

The magnetization of a given state is defined as

$$M(\sigma) = \eta \sum_i \sigma_i,$$

with η being some scalar. As we did with energy, we estimated $\langle M_T \rangle$ with its sample average $\langle \bar{M}_T \rangle$. Furthermore, we assume $\eta = 1$ in our analysis, as we are only concerned with the system's expected magnetization rather than its exact value at different temperatures.

Since we also want to determine if the material we are modeling will undergo a phase transition, we must compute the heat capacity C_T and susceptibility $\mathcal{X}_T$, since high variations in these values typically indicate a transition. They are defined as

$$C_T = \frac{\langle E_T^2 \rangle - \langle E_T \rangle^2}{kT^2},$$

$$\mathcal{X}_T = \frac{\langle M_T^2 \rangle - \langle M_T \rangle^2}{kT^2}.$$

These three quantities, alongside the energy of the system, are common macro quantities of the system.

3 Model Validation and Results

We validate the model by using the two-dimensional square grid model to find common observables and graph them. By comparing these results to sample graphs and known theoretical quantities, the model is shown to be accurate.

We first run multiple simulations of the model, and graph its energy per lattice point against the iteration of the simulation. We use a 30 x 30 lattice with 60% of the spins being candidates for reversal. We disregard the first 200 of 500 total iterations to allow the system to reach an equilibrium no matter what its initial random state may be, as shown by the vertical bar at k = 200.

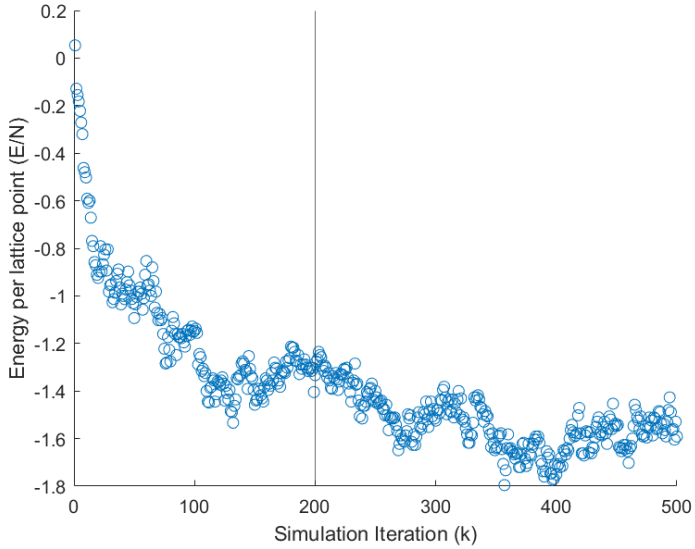


Figure 1: 30 x 30 simulated annealing with a temperature = 2

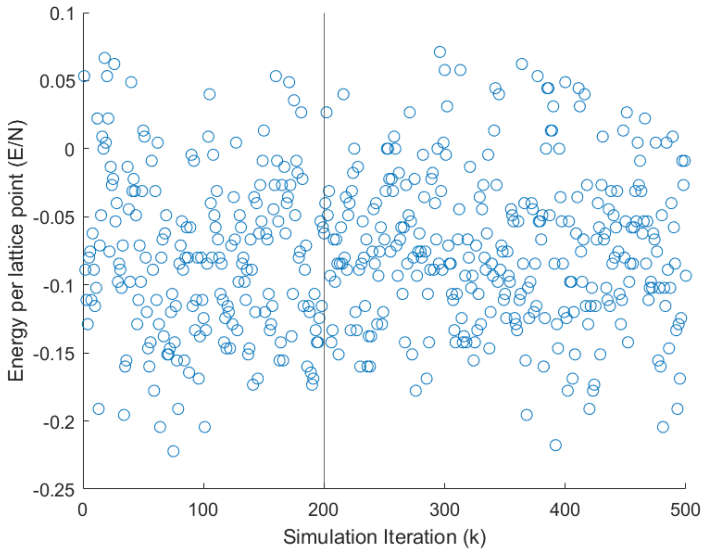


Figure 2: 30 x 30 simulated annealing with a temperature = 4

As Figures 1 and 2 show, the model eventually reaches an equilibrium energy per lattice point of approximately -1.5 when the temperature is equal to two, and -0.1 when the temperature is equal to four. As different temperatures induce different equilibrium states, this behavior is consistent with what we expect. Additionally, sample graphs as shown in Figure 3 show the same approximate behavior and trend towards the same equilibrium energies per lattice point that our model does, providing additional confidence in the model's validity.

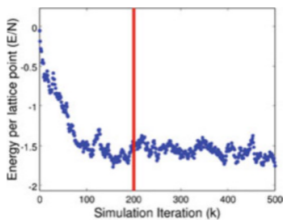


Fig. 12.11 The first 200 iterations of simulated annealing are discarded to let the system equilibrate from its random initiation. The lattice is 30×30 and 60% of the spins are candidates for reversal. The temperature is 2.

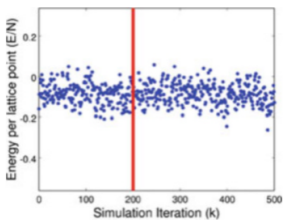


Fig. 12.12 The first 200 iterations of simulated annealing are discarded to ensure reasonable equilibration even though the system randomly starts near equilibrium. The lattice is 30×30 and 60% of the spins are candidates for reversal. The temperature is 4.

Figure 3: Sample simulated annealing

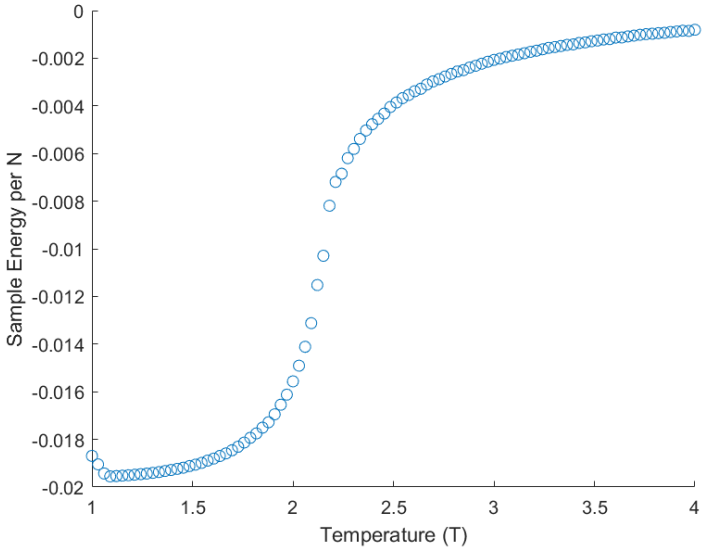


Figure 4: Simulated results estimating $\langle E_T \rangle / N$

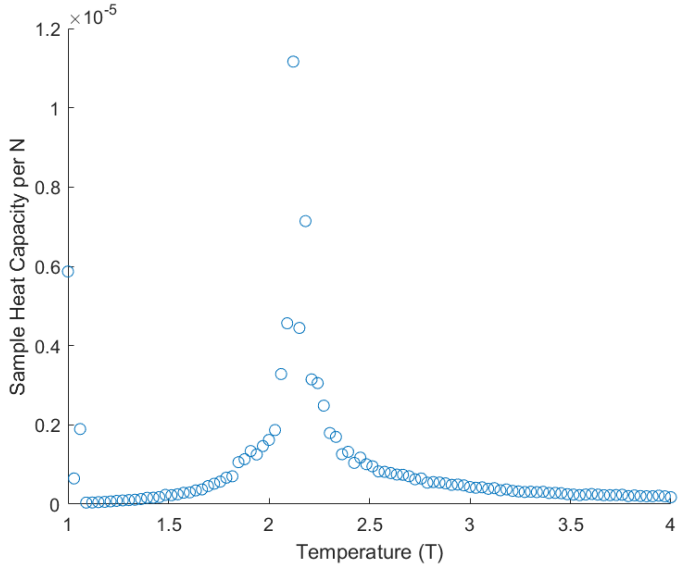


Figure 5: Simulated results estimating C_T/N

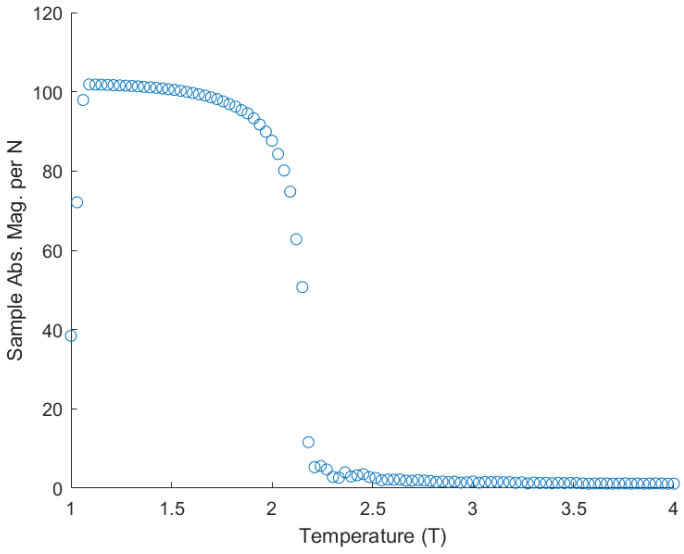


Figure 6: Simulated results estimating $\langle |M_T| \rangle / N$

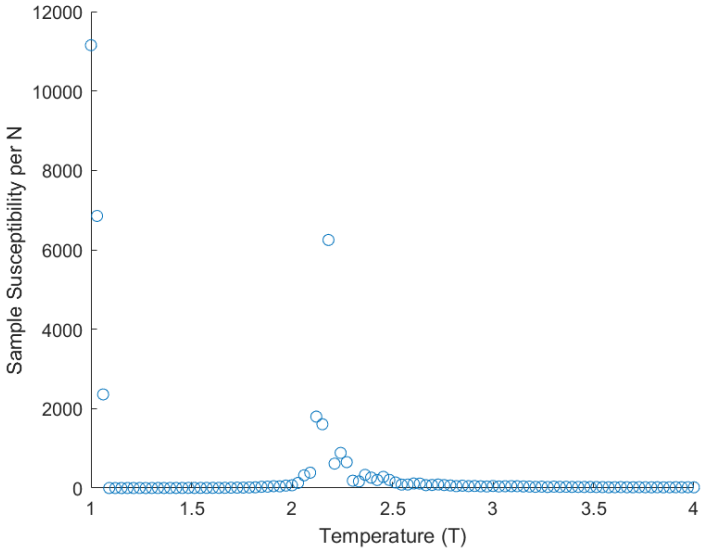


Figure 7: Simulated results estimating χ_T/N

We can see in Figures 4, 5, 6, and 7 that the material's properties experience a significant change around $T = 2.1$, at which the material achieves its highest heat capacity and susceptibility. What follows is a significant increase in energy and a rapid decline of magnetism. Because of this behavior, we can assert that the Curie temperature is at or around $T = 2.1$. It is proven in [1] that the true Curie temperature T_c is approximately 2.269, showing that our model is correctly approximating the Curie temperature accounting for error.

We also perform a linear regression on this model to approximate the critical exponents α, β, and γ, of the model, where

$$\begin{aligned} |M_t| &= |T - T_c|^{\beta} \approx N^{-\beta}, \\ |C_t| &= |T - T_c|^{\alpha} \approx N^{\alpha}, \\ |M_t| &= |T - T_c|^{\gamma} \approx N^{\gamma}. \end{aligned}$$

By using linear regression, these approximations can be obtained. For example, the statistical model $|M_T| = w * N^{-\beta} \Leftrightarrow ln(|M_T|) = ln(w) - \beta * ln(N)$ can be used to predict β. We simulate solutions for lattice dimensions

$N = 4, 5, 6, ..., 20$, with a temperature range of $2.15 \leq T \leq 2.30$, using 500 initiation iterates and 5000 sample iterates. The results of this linear regression can be seen in Figures 8, 9, and 10.

The theoretical values are $\alpha = 0.5, \beta = .125$, and $\gamma = 1.750$, while the predicted values are $\alpha = -1.2722, \beta = -1.3918$, and $\gamma = 2.8911$. These approximations thusly have a large margin of error compared to the theoretical value, and do not seem accurate enough to use in any real scientific context. This may be due to many factors in the model, such as the multiple instances of inherent randomness. However, as the model has already been verified through the simulations estimating the macro quantities, and this is an appropriate use of linear regression to approximate these critical exponents, this was deemed an acceptable amount of error for the scope of the project.

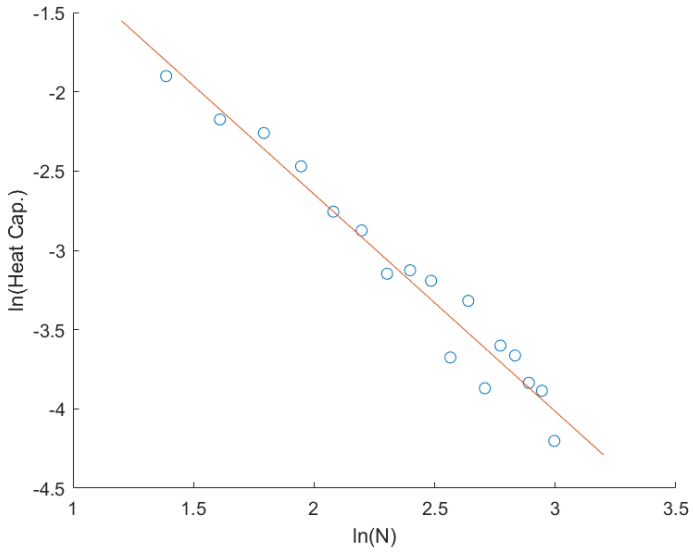


Figure 8: Predicting α with regression

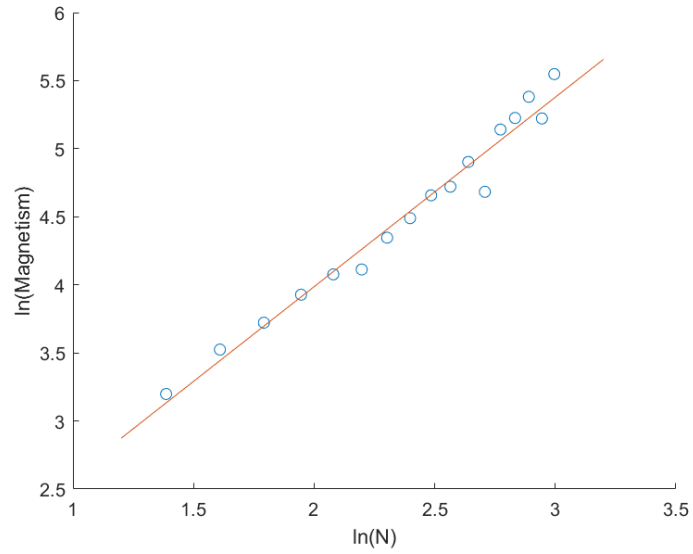


Figure 9: Predicting β with regression

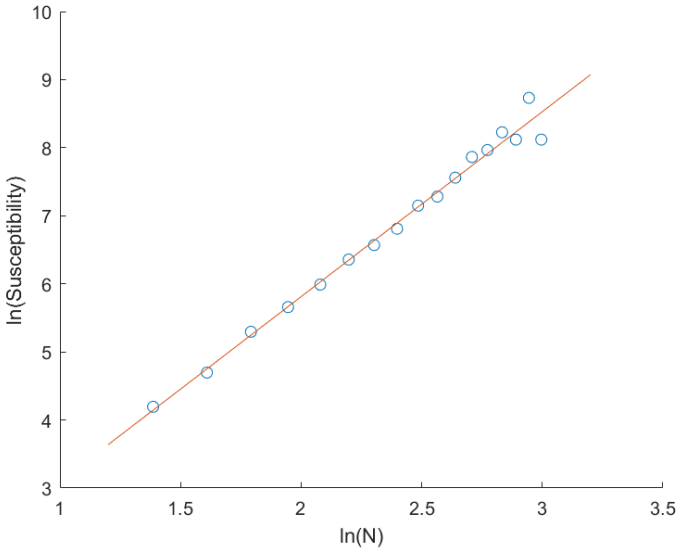


Figure 10: Predicting γ with regression

4 Experiments

The square lattice is naturally the easiest to both compute and visualize due to its simplicity and close correlation to a computer's matrix operations. However, not all atomic structures follow this pattern. We carry out two computational experiments using different, unique lattice structures.

Since the models are still run using a computer and thus matrices, the alternate lattice structures need to be converted into an equivalent square lattice with different neighbors. The two alternate geometries chosen are triangular and hexagonal. Figures 11 and 12 show both the original and converted lattice for each alternate geometry.

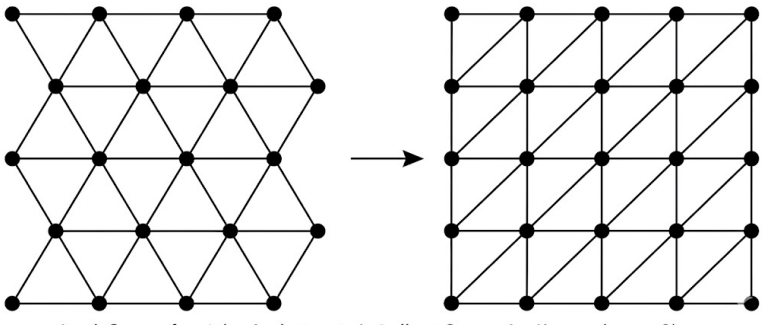


Figure 11: Triangle Lattice Diagram

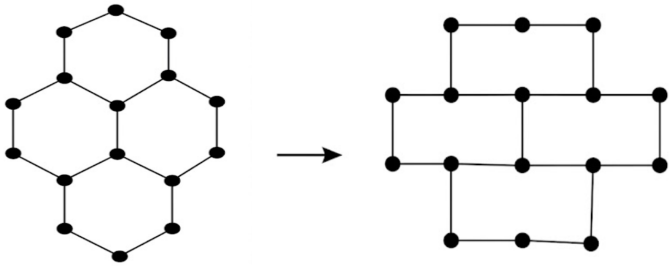


Figure 12: Hexagon Lattice Diagram

Our first experiment uses the triangular neighbor geometry. Figures 13, 14, 15, and 16 show the simulated results of estimating $< E_T > /N$, C_T/N, $< |M_T| > /N$, and χ_T/N respectively.

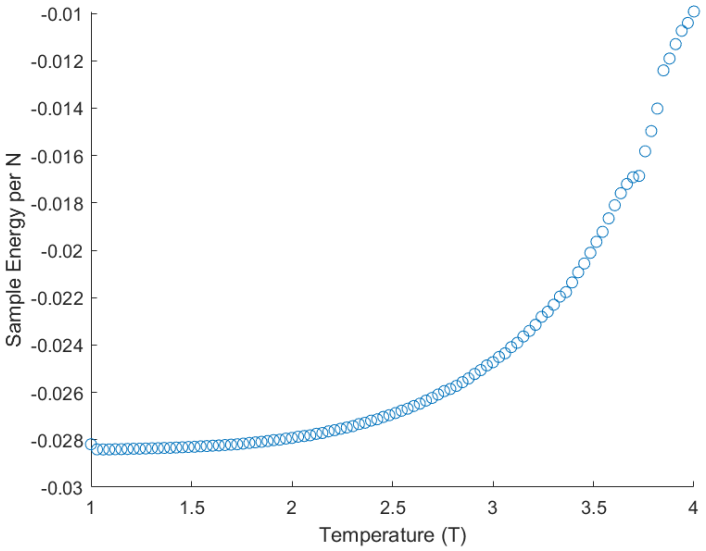


Figure 13: Simulated results estimating $< E_T > /N$

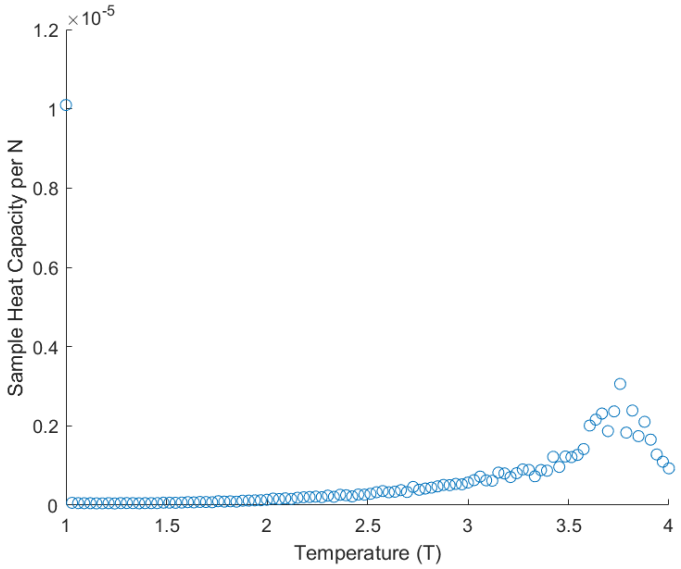


Figure 14: Simulated results estimating C_T/N

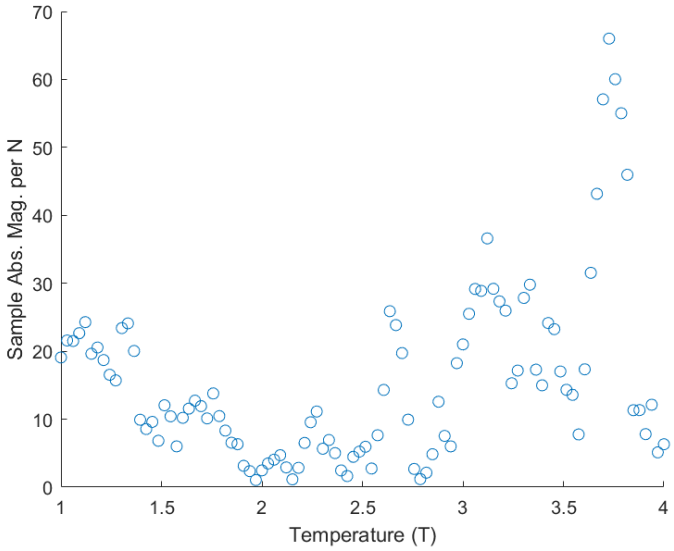


Figure 15: Simulated results estimating $< |M_T| > /N$

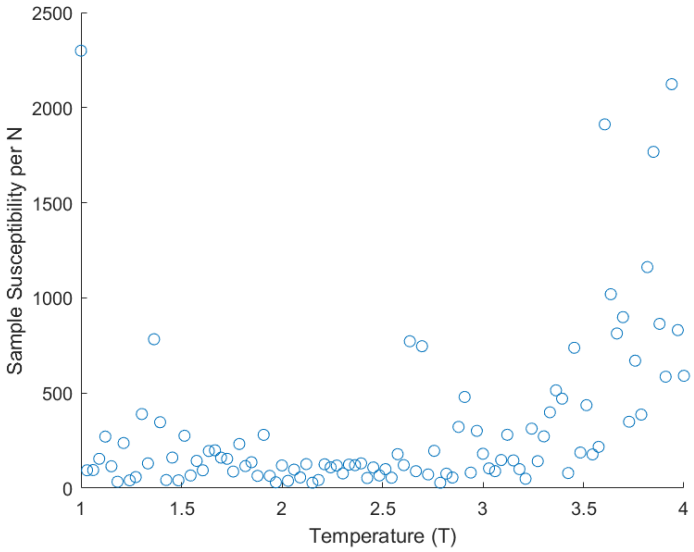


Figure 16: Simulated results estimating χ_T/N

As these figures show, there is a notable change in behavior between T = 3.5 and T = 4, which indicates that the true Curie Temperature for the model is within this interval. We calculate the temperature that gives the highest heat capacity across multiple runs of the model to obtain a more specific interval that is likely to contain the Curie Temperature.

These runs indicate that the Curie temperature is between T = 3.5 and T = 3.8, as all sample maximums fall within this interval. Therefore, this interval is used to carry out linear regressions to approximate the heat capacity, magnetism, and susceptibility of this model, as seen in Figures 17, 18, and 19.

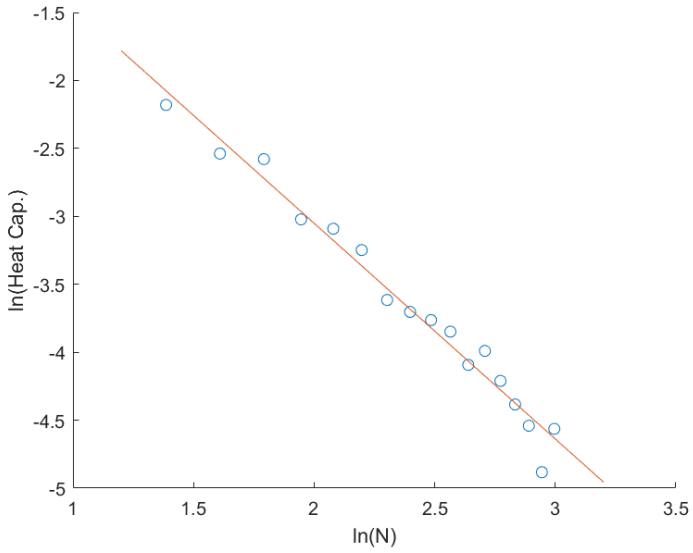


Figure 17: Predicting α with regression

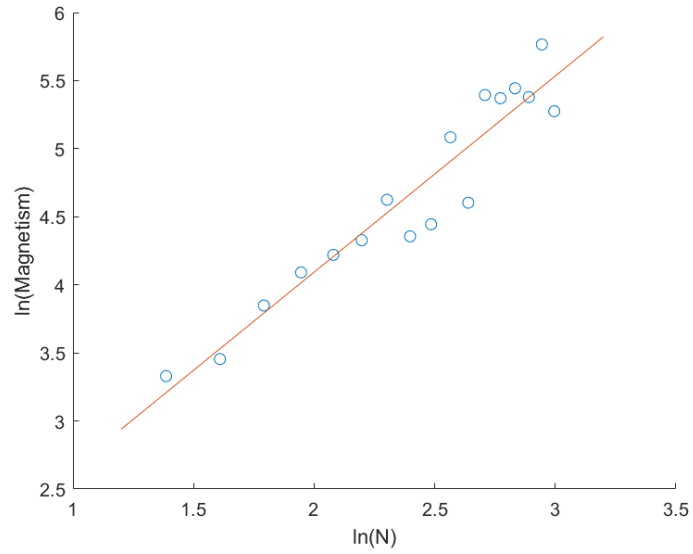


Figure 18: Predicting β with regression

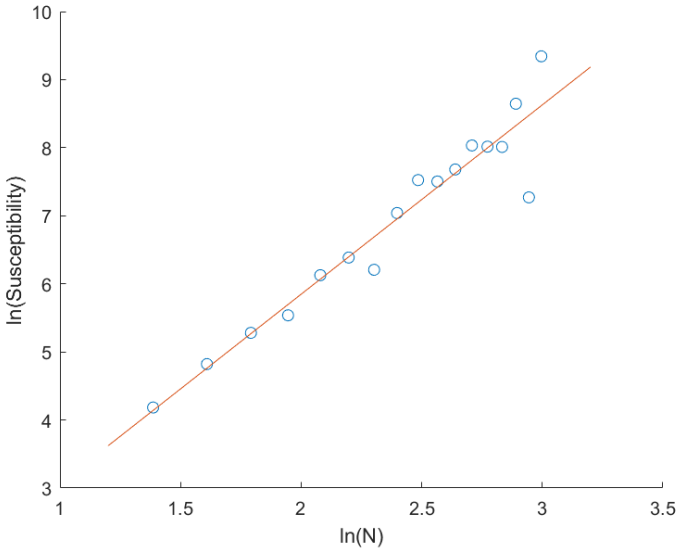


Figure 19: Predicting γ with regression

These linear regressions indicate that $\alpha = -1.4352$, $\beta = -1.5477$, and $\gamma = 2.7203$ for this model.

Our next experiment uses the hexagonal neighbor geometry. Figures 20, 21, 22, and 23 show the simulated results of estimating $< E_T > /N$, C_T/N, $< |M_T| > /N$, and χ_T/N respectively.

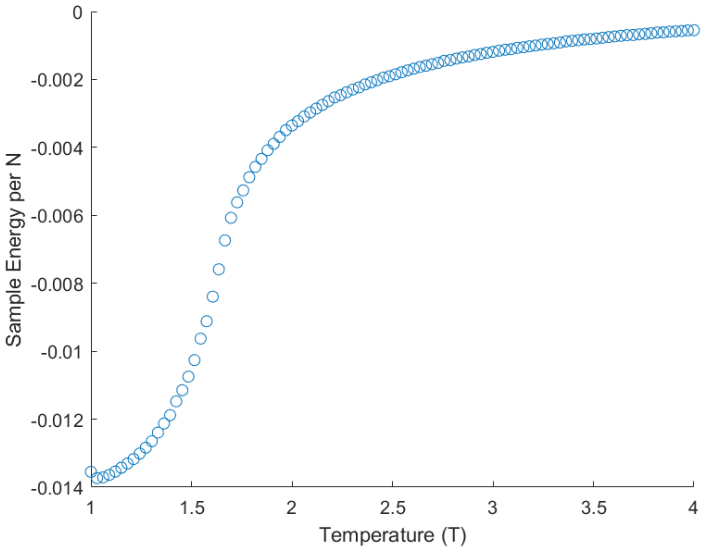


Figure 20: Simulated results estimating $< E_T > /N$

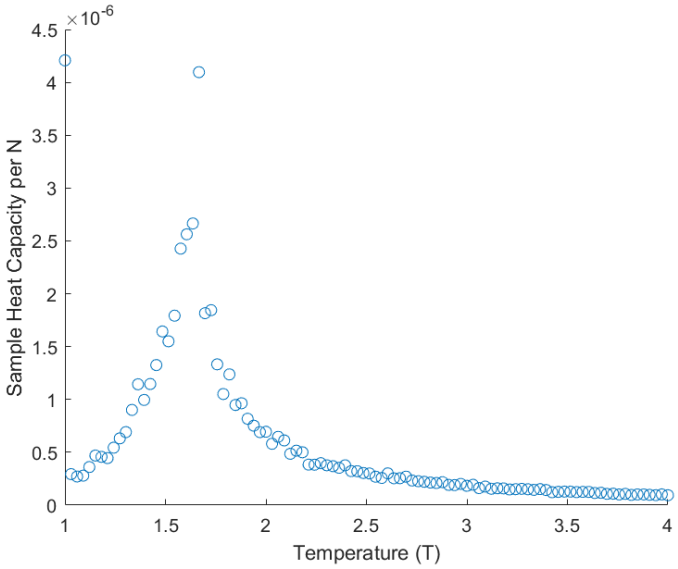


Figure 21: Simulated results estimating C_T/N

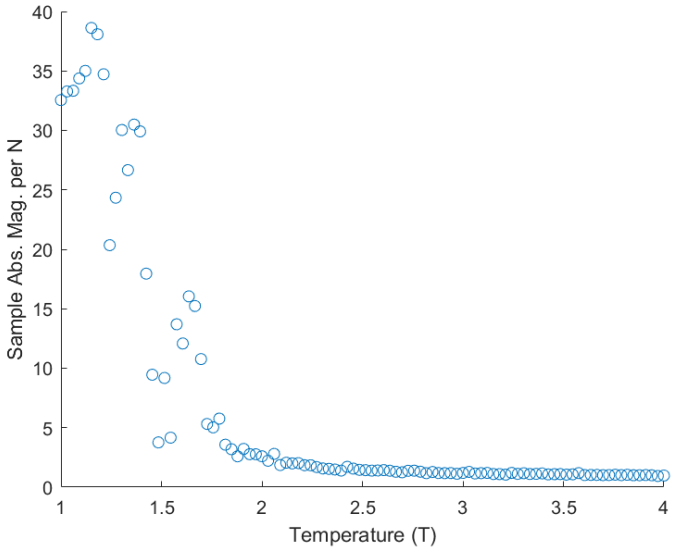


Figure 22: Simulated results estimating $< |M_T| > /N$

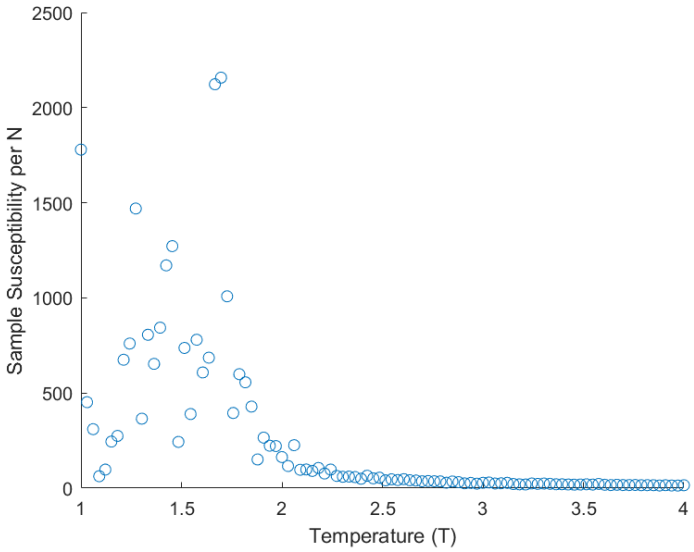


Figure 23: Simulated results estimating χ_T/N

These figures also show a notable change in behavior, but between T = 1.5 and T = 2 instead of T = 3.5 and T = 4. This indicates that the true Curie Temperature for this model is within the new interval. We calculate the temperature that gives the highest heat capacity across multiple runs of the model to obtain a more specific interval that is likely to contain the Curie Temperature.

These runs indicate the Curie temperature is between T = 1.5 and T = 1.8, as all sample maximums fall within this interval. Therefore, this interval is used to carry out linear regressions to approximate the heat capacity, magnetism, and susceptibility of this model, as seen in Figures 24, 25, and 26.

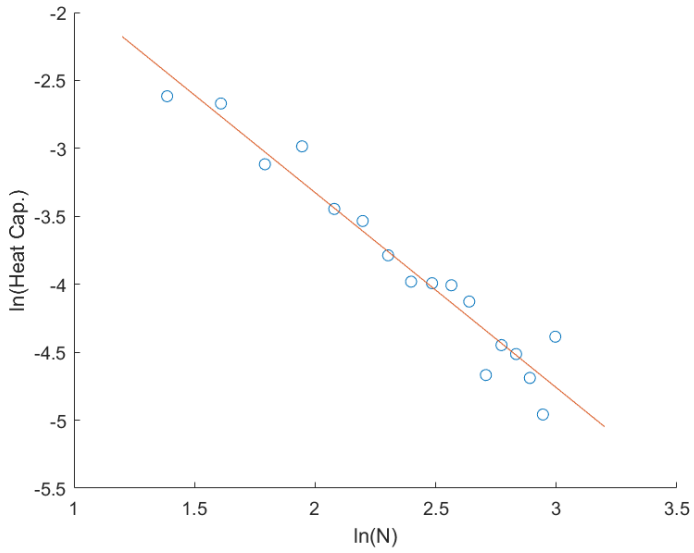


Figure 24: Predicting α with regression

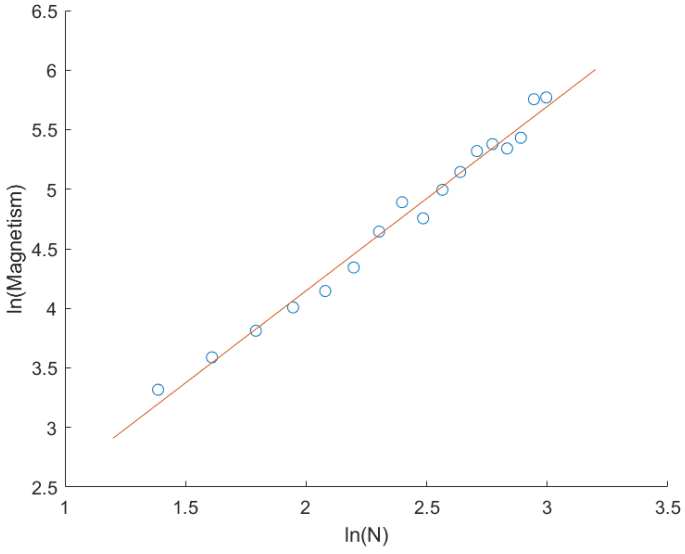


Figure 25: Predicting β with regression

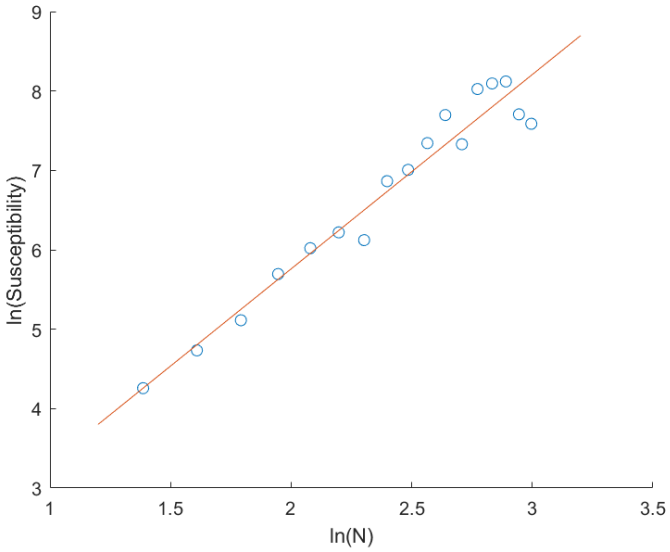


Figure 26: Predicting γ with regression

These linear regressions indicate that $\alpha = -1.5872$, $\beta = -1.4414$, and $\gamma = 2.7818$ for this model.

While this model's linear regression is not accurate at predicting the critical exponents, they both estimate very similar values for all three parameters, and so it is hard to judge how helpful or accurate one model is compared to another.

The differing Curie temperatures across the two experimental models are be reasonably explained by the differing number of connected neighbors in each model. Compared to the Curie Temperature of T = 2.269 with four neighbors in the square lattice, the hexagonal lattice has only three neighbors - leading to a lower Curie temperature - and the triangular lattice has six neighbors - leading to a much higher Curie temperature. This indicates that the number of neighbors that an individual point has in a model will influence its Curie temperature.

5 Code

Code for these models is written in MATLAB. All contributions can be found here: https://github.com/rhit-silkaijr/MA342Project5

6 Conclusion

The various figures and trends shown throughout this report show how this model is applicable to ferromagnetic materials and is able to calculate expected macroscopic properties from microscopic interactions such as Curie temperature, heat capacity, and susceptibility. This model is also shown to be consistent with other similar models such as the one found in Chapter 12 of Holder and Eichholz [1]. While this report is not specific to any real world material, further study can be done to experimentally find appropriate J, H, and η values that bridge the gap between computational modeling and the physical world.

References

[1] Allen Holder and Joseph Eichholz. *An Introduction to Computational Science*, volume 278. Springer Nature Switzerland AG, 2019.